

PROPOSED INTERNSHIP EXTENSION PROJECT

Simulation-to-Hardware Workflow for a 150 W LLC Resonant Converter

Low-voltage prototype. Industrial design method.

Measured correlation across circuit, magnetics, parasitics and thermal behavior.

RECOMMENDATION

Choose a hybrid: approximately 70% system and multiphysics design, 30% hardware validation.
If forced to choose only one original direction, practical-heavy is more portable in the job market;
the hybrid is stronger for both Lite-On relevance and portfolio evidence.

36-60 VDC

safe input window

12 V / 150 W

target output

100-250 kHz

switching range

10 weeks

baseline schedule

Prepared by Fransiskus Guntur

August 2026 | Mentor discussion draft | Lab equipment and Ansys licenses excluded from budget

01 Which direction is more beneficial?

Career value, internship fit and execution risk point to a hybrid scope.

Short answer: practical hardware experience is the stronger standalone signal for power-electronics jobs. However, your current Lite-On assignment is simulation-workflow oriented, and a pure hardware build may be harder to approve. A low-voltage LLC prototype used to validate an Ansys workflow captures both benefits.

Criterion	1. Practical-heavy LLC	2. Simulation-heavy PSU	Recommended hybrid
Hiring signal	Strong: schematic, PCB, bring-up, measurements	Moderate: analysis depth, but no proof of build/debug	Strongest: design plus measured model error
Fit to Lite-On power R&D;	High, but may compete for lab/supervision	High, especially with magnetics and workflow automation	Very high: converter, magnetics, simulation and test
Safety / access	Good below 60 VDC; high current still needs discipline	Best; no power-stage exposure	Good: SELV input and staged current-limited bring-up
Portfolio credibility	High if closed-loop and characterized	High only if reusable and correlated to real data	Highest: reproducible workflow + prototype
Main failure mode	Board works, but little engineering analysis	Beautiful plots with uncertain real-world accuracy	Scope creep; controlled by stage gates

Why this matches the current Taiwan job signal

Representative Lite-On power-hardware listings explicitly combine power-product development, magnetic-component calculation/design, design proposals, switching-topology knowledge, test methods and tools such as PSIM/SIMPLIS/Mathcad. Broader Taiwan postings similarly combine AC/DC or DC/DC design, PCB layout, magnetics, simulation, prototype debugging, thermal tests and EMI/EMC. The employable unit is therefore the complete design loop, not simulation or soldering in isolation. [1-3]

Decision rule

- If only one path is allowed: choose the low-voltage LLC hardware path.
- For the extension proposal: present the hybrid as a simulation-workflow project with a controlled hardware validation stage.
- Do not propose low-voltage PFC: at SELV input it becomes a boost-converter demonstrator and no longer exercises real AC-line power factor, rectifier behavior, hold-up, common-mode EMI or mains protection.

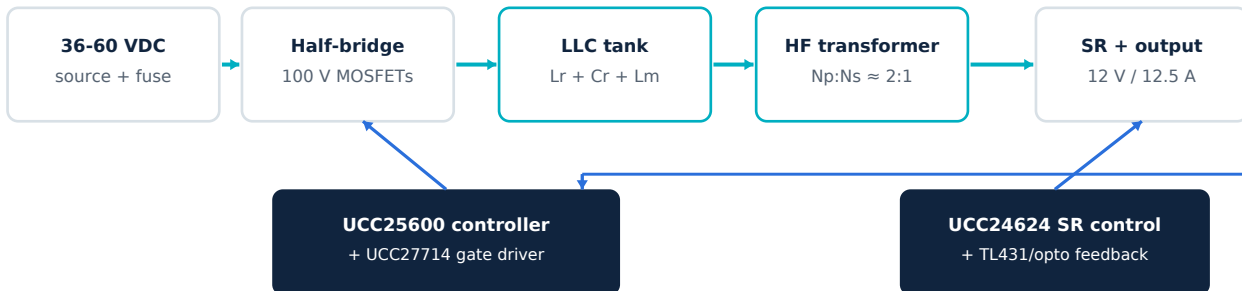
02 Recommended project definition

A bounded converter project that produces both an engineering asset and a demonstrable prototype.

Project title

Simulation-to-Hardware Correlation Workflow for a 150 W, 36-60 VDC to 12 VDC LLC Resonant Converter

System architecture



Engineering question

How accurately can an analytical + Ansys workflow predict the operating frequency range, resonant current, soft-switching boundary, transformer inductances/loss, PCB parasitics, efficiency and steady-state temperature of a real LLC converter?

Core scope (must finish)

FHA-based design; closed-loop analog control; custom transformer and resonant inductor; system switching model; Maxwell magnetic model; PCB parasitic extraction; first-pass thermal model; PCB bring-up; efficiency, waveform and temperature correlation.

Stretch scope (only after baseline)

Synchronous rectification optimization; automated parameter sweep/DOE; reduced-order magnetic model back-annotated into the system model; 12 V load-step optimization; comparison of two winding arrangements.

Explicit exclusions

- No direct AC mains input, no production safety certification claim, and no attempt to reproduce a confidential Lite-On platform.
- No full EMI compliance campaign in the baseline; only switching-node ringing and near-field / conducted pre-scan if equipment is available.
- No custom digital LLC control in the baseline. It would increase firmware and protection risk before the power stage is validated.

03 Target specification and baseline design

Targets are deliberately ambitious enough to be meaningful but bounded enough for a 10-week extension.

Parameter	Target	Reason / measurement
Input	36-60 VDC; 48 V nominal	Current-limited bench supply; no mains stage
Output	12.0 V, 0-12.5 A, 150 W	Electronic load; staged bring-up at 20/50/100/150 W
Topology	Half-bridge LLC, isolated	Frequency modulation; center-tapped SR secondary
Frequency	100-250 kHz; $f_r \approx 150$ kHz	Measure resonant frequency and ZVS window
Efficiency	$\geq 92\%$ at full load; peak $\geq 94\%$	Four-wire DC power measurement; report uncertainty
Regulation	$\pm 2\%$ over line/load	36/48/60 V $\times$ 10/50/100% load matrix
Ripple	< 150 mVpp, 20 MHz bandwidth	Short ground spring; define bandwidth in report
Transient	10 $\leftrightarrow$ 90% load: $< 5\%$ deviation, < 2 ms recovery	Electronic-load dynamic mode
Thermal	$< 100^\circ\text{C}$ hottest semiconductor; $< 90^\circ\text{C}$ core at 25°C ambient	Thermocouples + thermal camera after steady state
Protection	Input fuse, UVLO, OVP, OCP, thermal shutdown	Hard current limit independent of feedback loop

Starting-point electrical design - values must be finalized by gain analysis and simulation

Block	Baseline selection	Purpose / note
Controller	TI UCC25600	40-350 kHz LLC frequency controller; simple external bias and manageable baseline. [9]
Gate drive	TI UCC27714	4 A high-/low-side driver; electrically over-rated but robust and reusable. [10]
Primary FETs	100 V silicon MOSFETs, ≤ 5 m Ω	Adequate margin for a 60 V bus plus ringing; D2PAK / LFPACK footprint.
Resonant tank	$L_r \approx 2.2$ μH ; $C_r \approx 510$ nF; $L_m \approx 13$ μH	$f_r \approx 150$ kHz; $L_m/L_r \approx 6$. Values are design seeds, not frozen BOM.
Transformer	PQ32/30 or equivalent; $N_p:N_s \approx 2:1$	Gapped ferrite; litz/foil secondary; winding arrangement is a DOE variable.
Rectification	TI UCC24624 + 40 V SR MOSFETs	Dual-channel SR controller for LLC; baseline can first power up with diodes. [11]
PCB	4 layers, 2 oz outer copper preferred	Short commutation loops, Kelvin sensing, separated control return, test points.

04 Simulation and validation workflow

Each model must answer a design decision and be checked against a bench quantity.

Layer	Tool / model	Outputs	Correlation target
1. Analytical	FHA equations + Python/Mathcad	Gain curves, Q, Lm/Lr, frequency window, stress estimates	Explains trends; establishes feasible design space
2. Switching system	Ansys Twin Builder/Simplorer or SIMPLIS/PLECS	Startup, regulation, ZVS boundary, RMS/peak currents, loss inputs	fr $\pm 5\%$; key RMS currents $\pm 10\%$; efficiency ± 1.5 points
3. Magnetics	Ansys Maxwell 2D/3D	Lm, leakage, flux density, AC copper loss, core-loss map	Lm/leakage $\pm 10\%$; hotspot temperature $\pm 10^\circ\text{C}$
4. Parasitics	Ansys Q3D Extractor / SIwave	Commutation-loop L/R, winding capacitance proxy, SPICE RLCG	Drain overshoot/ringing frequency $\pm 15\%$
5. Thermal	Ansys Icepak	MOSFET, winding and core temperature; airflow sensitivity	Steady-state component temperature $\pm 10^\circ\text{C}$
6. Back-annotation	Measured parameters returned to system model	Updated waveforms and loss split; error table before/after	Document which mismatch came from model vs measurement

Why these tools are defensible

- Maxwell supports magnetic flux, saturation, leakage inductance and loss analysis for magnetic components. [4, 8]
- Q3D extracts frequency-dependent R, L, C and G and can export equivalent circuit models for power-converter interconnects. [5]
- Icepak predicts airflow, temperature and heat transfer in PCBs and power electronics. [6]
- Ansys explicitly positions the integrated workflow across converters, semiconductor loss, thermal behavior and EMI/EMC. [7, 8]

The portfolio artifact that matters

A one-page correlation matrix: predicted $\rightarrow$ measured $\rightarrow$ error $\rightarrow$ root cause $\rightarrow$ model update. This is more credible than a collection of contour plots and more useful to Lite-On than a one-off low-voltage board.

05 Ten-week learning and execution roadmap

The schedule uses stage gates so the project remains useful even if the stretch hardware is delayed.

Weeks	Learn	Build / analyze	Exit gate
1	LLC operating regions; FHA; ZVS/ZCS; gain sensitivity; RMS stress	Freeze requirements and test matrix; derive initial n , L_r , C_r , L_m	Gain range covers 36-60 V with margin
2	Frequency control; feedback; startup and protection	Ideal switching model; controller behavioral model; lossless checks	Stable regulation in line/load corners
3	Ferrite B-H behavior; Steinmetz limits; skin/proximity effect; fringing	Core selection, turns, gap, conductor geometry; Maxwell baseline	Bmax and loss targets met; build drawing released
4	MOSFET switching/conduction loss; SR timing; capacitor RMS	Device selection, loss budget, thermal source map, schematic review	Worst-case ratings and protections reviewed
5	High-di/dt layout; Kelvin paths; probe loops; parasitic resonance	PCB layout + Q3D loop extraction; update switching model	Layout review passed; PCB ordered
6	Electrothermal assumptions; boundary conditions; uncertainty	Icepak first pass; build magnetics and test control/gate-drive sections	Thermal margin and bring-up checklist approved
7	Safe bring-up; differential probing; fault isolation	20 W $\rightarrow$ 50 W open/closed-loop bring-up; capture tank and gate waveforms	No abnormal heating; stable 36/48/60 V operation
8	Efficiency measurement; uncertainty; dynamic loading	100 W $\rightarrow$ 150 W; line/load matrix; thermal steady state	Baseline performance dataset complete
9	Model calibration vs overfitting; sensitivity and DOE	Back-annotate measured L/R/C and losses; run winding/air-gap DOE	Correlation errors within target or explained
10	Design review and technical storytelling	Release model template, scripts, BOM, test SOP, final report and demo	Another engineer can reproduce the workflow

If the extension is shorter

6 weeks: stop after magnetics sample + 50 W correlation. 8 weeks: complete 150 W baseline without DOE. 10-12 weeks: complete the full workflow and one design optimization.

06 Budget, resources and risk control

Costs are prototype estimates in Taiwan, August 2026; company lab assets are excluded.

Item	Estimate (NTD)	Included
PCB fabrication	1,500-3,000	4-layer prototypes, shipping, stencil if needed
Power semiconductors + drivers	1,200-2,200	Primary MOSFETs, SR MOSFETs, UCC25600, UCC27714, UCC24624, spares
Magnetics materials	900-1,800	Cores, bobbins, litz/foil, insulation, resonant inductor materials, two variants
Passives, sensing, protection	1,200-2,000	Film capacitors, electrolytics, shunts, optocoupler, fuse, connectors
Mechanical / thermal	500-1,000	Heatsinks, fan options, interface pads, fixtures
Rework and second revision reserve	1,500-3,000	Replacement parts and one targeted PCB spin
Total prototype budget	6,800-13,000	Excludes Ansys, oscilloscope, probes, DC source, e-load, LCR/impedance analyzer, thermal camera

Minimum company resources requested

- 36-60 V / ≥ 5 A current-limited DC source; ≥ 200 W electronic load; 4-channel oscilloscope with differential voltage and current probes.
- LCR or impedance analyzer; thermal camera + thermocouples; solder/rework station; supervised bench area.
- Ansys Electronics Desktop access: Twin Builder/Simplorer, Maxwell, Q3D or SIwave, and Icepak; one named technical reviewer for weekly 30-minute design gates.

Risk	Control	Fallback that preserves value
Magnetics iteration delay	Order two core/bobbin sets; characterize before PCB power-up	Use measured magnetics with external wiring at 50 W
Unstable LLC loop	Validate plant model; conservative compensation; current-limited startup	Demonstrate regulated 50 W baseline; keep full-power open-loop sweep
SR shoot-through	Initial diode rectifier bring-up; add SR only after waveforms are proven	Retain diode baseline and lower efficiency target
Tool-access mismatch	Confirm licensed modules in week 0	Use available Ansys tools for magnetics/thermal and PLECS/SIMPLIS for system layer
Scope creep	One baseline topology; DOE only after 150 W dataset	Deliver reusable workflow and 50 W hardware correlation

07 Deliverables and mentor decision

The proposal is successful if it leaves Lite-On with a reusable method, not just a student demonstration board.

Technical package

1. Requirements and design calculations
2. Schematic, PCB and BOM
3. Transformer/inductor build drawings
4. Parameterized Ansys models
5. Test plan, raw data and uncertainty notes
6. Correlation/error report
7. Final demonstration and design review

Resume-level outcome

“Developed and validated a 150 W LLC resonant converter workflow spanning analytical design, system simulation, Ansys Maxwell magnetics, Q3D parasitics and Icepak thermal analysis; built a 36-60 VDC prototype and correlated resonant frequency, losses and temperature against bench measurements.”

Approval requested from mentor

Decision	Proposed answer
Project direction	Approve simulation-led LLC design with low-voltage hardware correlation
Baseline duration	10 weeks; reduce to 6-8 weeks using the defined fallback scope
Review cadence	30-minute weekly design gate; schematic/layout and bring-up gates require sign-off
Lab scope	≤60 VDC input only; staged power levels; supervised first energization
Tool availability	Confirm Maxwell + Q3D/Siwave + Icepak + system-level simulator before week 1
Business/IP boundary	Use generic specifications and public components unless a sanitized internal design is explicitly provided

One-sentence pitch to the mentor

“I propose building a reusable simulation-to-hardware workflow around a safe 150 W LLC converter, so the extension produces both a validated Ansys method and real power-supply design experience without requiring direct work on the high-voltage PFC front end.”

Final recommendation

Approve the hybrid. It is the most defensible extension for your current role and the strongest portfolio signal: topology selection, magnetics, modeling, PCB/layout thinking, protection, measurement, debugging and honest correlation in one bounded project.

08 Sources and technical anchors

Accessed 21 August 2026. Product values in this proposal remain subject to final component availability and design review.

1	Lite-On power hardware engineer listing - product development, magnetics, design proposals, topology and test methods. https://www.104.com.tw/job/8iewk
2	Lite-On company recruiting page - current power-electronics hiring context. https://www.104.com.tw/company/aqae7i0
3	104 Taiwan power-engineer search - representative requirements across AC/DC, DC/DC, PCB, magnetics, simulation, prototype and EMI/EMC. https://www.104.com.tw/jobs/search/?jobcat=2008001013
4	Ansys Maxwell - electromagnetic field solver for electrical power conversion and magnetic devices. https://www.ansys.com/products/electronics/ansys-maxwell
5	Ansys Q3D Extractor - frequency-dependent RLCG extraction and equivalent circuit generation. https://www.ansys.com/products/electronics/ansys-q3d-extractor
6	Ansys Icepak - electronics cooling and temperature/airflow prediction. https://www.ansys.com/products/electronics/ansys-icepak
7	Ansys power electronics workflow overview - system-level converters, loss, thermal and EMI/EMC analysis. https://www.ansys.com/applications/power-electronics
8	Ansys magnetic component design course - flux, saturation, leakage inductance and loss in Maxwell/Simplorer. https://learninghub.ansys.com/learn/course/external/view/classroom/2547/magnetic-component-design-using-ansys-maxwell-software
9	Texas Instruments UCC25600 - 40-350 kHz high-performance LLC resonant controller. https://www.ti.com/product/UCC25600
10	Texas Instruments UCC27714 - 600 V, 4 A high-side/low-side gate driver. https://www.ti.com/product/UCC27714
11	Texas Instruments UCC24624 - dual-channel synchronous rectifier controller for LLC. https://www.ti.com/product/UCC24624
12	Texas Instruments digital LLC reference - complete converter control and >90% efficiency context. https://www.ti.com/tool/TIDM-RESLLC-DCDC

Engineering note: The proposed resonant-tank values, transformer ratio and component classes are credible starting points for scoping and budgeting, not released production values. Final values follow from the week-1 gain analysis, verified device models, magnetic characterization and mentor design review.